E69: High-Regularity L^∞ Bound for the Corrected Regularized Curvature Operator

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Abstract

We prove that the corrected complex curvature operator is essentially bounded whenever its input lies in $H^s(\mathbb{R}^n)$ with $s > n/2 + 2$ and the regularization parameter ε is positive. The regularized reciprocal is bounded globally, including at zeros of the field. The estimate is explicit up to the Sobolev embedding constant and does not require the spatial dimension to be at most three.

1 Statement

For $\varepsilon > 0$ and $\alpha \in \mathbb{R}$, define, for $z \in \mathbb{C}$,

$$R_\varepsilon(z) := \frac{\overline{z}}{|z|^2 + \varepsilon^2 \exp(-2\alpha|z|^2)}, \tag{1}$$

and, for a complex-valued field ψ,

$$\Theta_\varepsilon[\psi] := -(\Delta\psi)\, R_\varepsilon(\psi) = -\Delta\psi\, \frac{\overline{\psi}}{|\psi|^2 + \varepsilon^2 \exp(-2\alpha|\psi|^2)}. \tag{2}$$

Theorem 1 (E69: bounded regularized curvature)**.** *Let $n \geq 1$, $\varepsilon > 0$, $\alpha \in \mathbb{R}$, and $s > n/2 + 2$. If $\psi \in H^s(\mathbb{R}^n; \mathbb{C})$, then*

$$\Theta_\varepsilon[\psi] \in L^\infty(\mathbb{R}^n; \mathbb{C}).$$

In fact, with $\alpha_+ := \max\{\alpha, 0\}$ and

$$M_{\varepsilon,\alpha} := \max\left\{1, \frac{e^{2\alpha_+}}{\varepsilon^2}\right\}, \tag{3}$$

there is a finite constant $C_{n,s} > 0$ independent of $\psi, \varepsilon, \alpha$ such that

$$\boxed{\|\Theta_\varepsilon[\psi]\|_{L^\infty(\mathbb{R}^n)} \leq C_{n,s} M_{\varepsilon,\alpha} \|\psi\|_{H^s(\mathbb{R}^n)}.} \tag{4}$$

2 Proof

Proof. We first estimate the Laplacian. With the unitary Fourier transform,

$$\widehat{\Delta\psi}(\xi) = -|\xi|^2 \widehat{\psi}(\xi).$$

Consequently,

$$\|\Delta\psi\|_{H^{s-2}}^2 = \int_{\mathbb{R}^n} (1+|\xi|^2)^{s-2} |\xi|^4 |\widehat{\psi}(\xi)|^2 \, d\xi \tag{5}$$

$$\leq \int_{\mathbb{R}^n} (1+|\xi|^2)^{s} |\widehat{\psi}(\xi)|^2 \, d\xi = \|\psi\|_{H^s}^2, \tag{6}$$

where we used $|\xi|^4 \leq (1+|\xi|^2)^2$. Thus

$$\|\Delta\psi\|_{H^{s-2}} \leq \|\psi\|_{H^s}. \tag{7}$$

Since $s-2 > n/2$, the Sobolev embedding $H^{s-2}(\mathbb{R}^n) \hookrightarrow L^\infty(\mathbb{R}^n)$ applies. Therefore

$$\|\Delta\psi\|_{L^\infty} \leq C_{n,s} \|\Delta\psi\|_{H^{s-2}} \leq C_{n,s} \|\psi\|_{H^s}. \tag{8}$$

For completeness, the embedding may be obtained by Cauchy–Schwarz: for $r = s-2 > n/2$,

$$\|\widehat{\Delta\psi}\|_{L^1} \leq \left(\int_{\mathbb{R}^n} (1+|\xi|^2)^{-r} \, d\xi \right)^{1/2} \|\Delta\psi\|_{H^r}, \tag{9}$$

and the integral is finite since its radial tail behaves as $\int_1^\infty \rho^{n-1-2r} \, d\rho$. Fourier inversion then yields (8).

Next we bound the reciprocal (1). For any $z \in \mathbb{C}$, the denominator

$$D_\varepsilon(z) := |z|^2 + \varepsilon^2 e^{-2\alpha|z|^2}$$

is strictly positive, because $\varepsilon > 0$. Put $t = |z| \geq 0$. If $0 \leq t \leq 1$, then $e^{-2\alpha t^2} \geq e^{-2\alpha_+}$, so

$$|R_\varepsilon(z)| = \frac{t}{t^2 + \varepsilon^2 e^{-2\alpha t^2}} \leq \frac{e^{2\alpha_+}}{\varepsilon^2}.$$

If $t \geq 1$, the denominator is at least t^2, and hence

$$|R_\varepsilon(z)| \leq \frac{t}{t^2} = \frac{1}{t} \leq 1.$$

Thus, uniformly in z,

$$\sup_{z \in \mathbb{C}} |R_\varepsilon(z)| \leq M_{\varepsilon,\alpha} < \infty. \tag{10}$$

In particular, $R_\varepsilon(0) = 0$, so nodes of ψ produce no singularity.

Finally, by (2), (8), and (10),

$$\begin{aligned} \|\Theta_\varepsilon[\psi]\|_{L^\infty} &\leq \|\Delta\psi\|_{L^\infty} \|R_\varepsilon(\psi)\|_{L^\infty} \\ &\leq C_{n,s} \|\psi\|_{H^s} M_{\varepsilon,\alpha}. \end{aligned}$$

This proves (4) and the asserted L^∞ membership. □

3 Scope and limitations

Remark 1 (Spatial dimensions)**.** *The sufficient requirement $s > n/2 + 2$ applies in every finite spatial dimension. It requires $s > 5/2, 3, 7/2, 4, 9/2$ for $n = 1, 2, 3, 4, 5$, respectively. By contrast, the embedding $H^2 \hookrightarrow L^\infty$ for $n \leq 3$ bounds ψ, not necessarily $\Delta\psi$; H^2 regularity alone does not yield Theorem 1.*

Remark 2 (Other domains)**.** *The same reasoning works on a flat torus, with the periodic Sobolev embedding, and on a bounded Lipschitz extension domain Ω when an $H^s(\Omega) \to L^\infty(\Omega)$ embedding for second-order derivatives is available (for example, under the usual suitable extension and regularity hypotheses). In that case the constant depends on Ω.*

Remark 3 (Fixed regularization)**.** *The bound is for fixed $\varepsilon > 0$ and does not imply a uniform estimate as $\varepsilon \downarrow 0$. Nor does it establish existence, uniqueness, or stability of a PDE flow involving Θ_ε.*

Remark 4 (Choice of regularizer)**.** *The claim uses the modulus-squared regularizer* (1)*. It does not extend automatically to the older complex denominator $\psi + \varepsilon e^{-\alpha|\psi|^2}$, which can vanish.*